Angles from intersecting chords, secants, and tangents

1

a $x = 103$

b $x = 123$

c $x = 95, y = 85$

d $x = 25$

e $x = 74$

f $x = 38$

g $x = 245, y = 65$

h $x = 60, y = 148$

i $x = 180, y = 45, z = 45$

2

a 110°

b 35°

c 4.88 units

3

a 114°

b 114°

c 33°

d 33°

e 66°

4

a 90°

b 10

c $5\sqrt{3}$

d 36°

e 72°

5

a 13.5°

b 53°

c 100°

d 50°

e 63.5°

Additional questions

6

a $m\overset{\frown}{AEB} = 266^\circ$

b $m\angle ACE = 40^\circ$

c $x = 40$

d $x = 64$

e $m\angle UVT = 114^\circ$

f $x = 10$

g $m\angle DEB = 81^\circ$

h $m\overset{\frown}{TU} = 130^\circ$

i $x = 43^\circ$

j $m\angle H = 52^\circ$

7

a $x = 7.25$

b $m\angle 1 = 33.75^\circ$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8

a $x = 20.6$ **b** $m\overset{\frown}{ADC} = 172^\circ$

9

a $m\overset{\frown}{EC} = 136^\circ$ **b** $m\angle BEC = 48^\circ$

10 $x = 12$

11

a $m\angle 1 = 28.75^\circ$ **b** $m\overset{\frown}{AB} = 91^\circ$

12

a 120° **b** 91°

13 Slices B, C, and D would fit. Slice D would not because the arc has a measure of 146°.

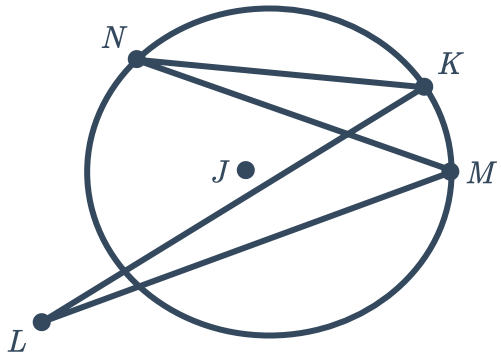
14

a $x = 6$ **b** $m\angle 1 = 49.5^\circ$ **c** $m\overset{\frown}{EDC} = 125^\circ$ **d** $m\angle 2 = 61.5^\circ$

15 $\angle KLM$ is formed by a secant and a tangent, so $m\overset{\frown}{KL} = 2x^\circ$. $\angle KJL$ intercepts $\overset{\frown}{KL}$, so $m\angle KJL = x^\circ$. Since the given triangle is isosceles we know that $m\angle KLJ = x^\circ$, and using triangle sum theorem we get $x = 45$. Since $x = 45$, $m\angle JKL = 90^\circ$, meaning it must be inscribed in a diameter. Therefore $\overline{JL}$ is a diameter of the circle.

16

a Model answer:



b $\angle KLM = \frac{1}{2}\angle N$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647